

Shubham Kumar

📞 9877883408 | ✉️ sshubhamk1@hotmail.com | 🌐 github.com/sshubhamk1 | 🔗 linkedin.com/in/sshubhamk1

Senior Backend Engineer specializing in scalable distributed systems, event-driven architectures, and low-latency backend platforms handling high-throughput real-time workloads

Skills

- Programming: Python, Go, Node.js, Java, SQL, Bash
- Backend & APIs: Django, WebSockets, REST APIs, Microservices, Django Rest Framework(DRF)
- Distributed Systems & Messaging: Kafka, Celery, Redis, Event-Driven Architecture, Asynchronous Processing, Fault Tolerance, High Availability, Scalability
- Databases & Storage: PostgreSQL, Elasticsearch, pgvector
- Cloud & Infrastructure: AWS, GCP, Docker, Nginx
- Systems Engineering: System Design, Performance Optimization, Observability, Load Testing, Debugging

Work Experience

Senior Backend Engineer – Collective Artists Network, Bangalore, IN Sep 2024 – Mar 2026

- Revamped personalized content delivery for **2M+ users** by designing an **embedding-based feed ranking system**.
- Increased platform engagement to **50K+ DAU** by designing a **scalable activity-driven leaderboard system**.
- Mentored engineers and established debugging, observability, and reliability best practices, reducing production incidents by 25%.

Independent Backend Engineer Nov 2023 – Aug 2024

- Improved production reliability and service availability by designing a microservices platform supporting zero-downtime deployments across **10+ microservices**.
- Enabled reliable real-time processing of **500+ stock** symbols by designing a fault-tolerant event streaming pipeline.
- Improved reliability of asynchronous workflows by building a scalable job orchestration system using Celery and Redis.

Senior Software Engineer – Radisys Corporation, Bangalore, IN Jul 2021 – Nov 2023

- Reduced boot time by **60%** by developing a fault-tolerant supervision framework with automated recovery mechanisms.
- Resolved critical production crashes through core dump analysis and Linux memory debugging.
- Led a team of 5 engineers, driving architecture decisions, code quality standards, and engineering best practices.

Projects

Real-Time Video Streaming Infrastructure: Built a video processing and adaptive streaming platform using S3/MinIO, FFmpeg, and queue-based parallel workers, improving concurrent upload throughput and reducing media delivery latency through chunk-based transcoding, HLS/DASH streaming, pre-signed URLs, and CDN-ready architecture.

Achievements

- Published **3 peer-reviewed research papers** on IoT malware detection, botnet architectures, and distributed security systems in IEEE and Springer publications.
- Participated in **100+ CTF competitions** focused on web security, reverse engineering, and Linux systems.
- Built and shipped functional prototypes across 10+ hackathons under aggressive development timelines.

Education

NITK Surathkal – M.Tech(Research), Computer Science & Engineering – Information Security Dec 2022
Lovely Professional University – B.Tech, Computer Science & Engineering May 2019